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Appendix

Data Dictionary of the Analytical Base Table (NYC_Taxi_AB.T.csv)

The data dictionary prepared by the group for this study is reproduced on the following pages.

Data Dictionary

NYC Yellow Taxi Fleet Deployment and Revenue Optimization — Analytical Base Table (NYC_Taxi_ABT.csv)

Purpose of this document

This data dictionary documents every column in the Analytical Base Table (NYC_Taxi_ABT.csv), which serves as the single shared source dataset for all three models in this study (XGBoost, SVR, and Neural Network). It distinguishes between raw fields carried over from the original TLC trip records and engineered fields created during ABT construction, and notes how each field relates to the study's target variable, predictors, or analytical questions.

Field categories

Raw/source fields: VendorID, tpep_pickup_datetime, tpep_dropoff_datetime, passenger_count, trip_distance, RatecodeID, store_and_fwd_flag, PULocationID, DOLocationID, payment_type, fare_amount, extra, mta_tax, tip_amount, tolls_amount, improvement_surcharge, total_amount, congestion_surcharge, Airport_fee, cbd_congestion_fee — carried directly from the original TLC Yellow Taxi trip record files.

Engineered fields: duration_min, speed_mph, pickup_hour, day_of_week, is_weekend, pickup_year, total_fees, net_revenue, net_revenue_per_hour, fee_impact_pct, fare_tier, trip_length_tier, congestion_level — calculated or derived during ABT construction to support the study's business question and modeling approach.

Column / Variable Name	Data Type	Description / Meaning	Possible Values / Categories
VendorID	Integer	Identifies which TPEP (Taxicab Passenger Enhancement Program) technology provider generated and transmitted this trip record to the TLC. This is an administrative/system field — it says nothing about trip quality or revenue, and is not used as a model input. Kept in the ABT only for traceability back to the source system.	1, 2, 6 (per official TLC vendor codes)
tpep_pickup_datetime	Datetime	The exact timestamp when the taxi meter was engaged, marking the start of the trip. This is the anchor field from which pickup_hour, day_of_week, is_weekend, and pickup_year are all derived. In the modeling notebook it is floored to the hour to form the zone-hour unit of analysis, which is what the chronological train/validation/test split is based on.	2025-01-01 00:00:17 to 2025-11-30 23:59:59
tpep_dropoff_datetime	Datetime	The exact timestamp when the meter was disengaged, marking the end of the trip. Used together with tpep_pickup_datetime to calculate duration_min. Not used directly as a model input, since a trip's own dropoff time would not be known in advance in a real deployment scenario.	2025-01-01 00:03:34 to 2025-12-01 00:29:37
passenger_count	Integer	Number of passengers reported in the vehicle for the trip, entered manually by the driver. A value of 0 is possible (e.g., unreported) and was checked during cleaning as a potential data-quality flag; the modeling notebook keeps only trips with at least one passenger.	0 to 9
trip_distance	Float	Distance covered during the trip, in miles, as recorded by the taxi meter (not straight-line/GPS distance). Used to compute speed_mph and to describe trip-length characteristics such as trip_length_tier.	0.11 to 99.06 miles
RatecodeID	Integer	Indicates which fare-calculation rule applied at the end of the trip. Standard trips use metered fare-and-distance pricing; other codes indicate flat or negotiated rates (e.g., airport trips), which can affect how fare_amount relates to trip_distance and duration. The modeling notebook keeps codes 1 to 6.	1 = Standard rate, 2 = JFK, 3 = Newark, 4 = Nassau/Westchester, 5 = Negotiated fare, 6 = Group ride, 99 = Unknown/Other

Column / Variable Name	Data Type	Description / Meaning	Possible Values / Categories
store_and_fwd_flag	String	Indicates whether the trip record was temporarily stored in the vehicle's memory because the vehicle had no live connection to the vendor's server before reaching the passenger's destination, and was later forwarded ("store and forward"). This is a data-transmission detail rather than a business measure, and is not used in modeling.	Y = record was store-and-forward, N = record was sent immediately (most trips)
PULocationID	Integer	The TLC Taxi Zone ID identifying where the trip began (pickup zone). This is the primary location key used to build the Zone x Date x Hour unit of analysis for modeling, and is what deployment recommendations (AQ1) are ultimately expressed in terms of.	1 to 263 (TLC Taxi Zone Lookup Table); 258 zones present
DOLocationID	Integer	The TLC Taxi Zone ID identifying where the trip ended (dropoff zone). Useful for understanding trip flow between zones, but is not used as a predictor for deployment decisions, since a trip's destination is not known at the time a driver is dispatched.	1 to 263 (TLC Taxi Zone Lookup Table)
payment_type	Integer	Code indicating how the passenger settled the fare. Included for completeness/possible EDA use (e.g., checking if payment method correlates with tipping behavior), but not used as a model predictor since it has no bearing on deployment strategy. Flex-fare trips (code 0) leave Airport_fee blank and record congestion_surcharge as 0 in the source data (see congestion_surcharge).	0 = Flex Fare, 1 = Credit card, 2 = Cash, 3 = No charge, 4 = Dispute
fare_amount	Float	The base time-and-distance fare calculated by the taxi meter, before extras, taxes, tolls, tips, and congestion-related fees are added. This is the starting point for total_amount.	0.01 to 1,334.02 (USD)
extra	Float	Miscellaneous surcharges not covered by other fee columns, such as rush-hour or overnight differential charges. For VendorID 1, this column also includes the congestion surcharge and CBD fee that are reported in their own columns. Excluded from the total_fees calculation used for net_revenue, since it is not one of the congestion/regulatory fees the study focuses on. Negative values are refund artifacts; the modeling notebook removes those trips.	-17.39 to 15.00 (USD)
mta_tax	Float	A fixed state-mandated tax automatically applied to metered fares. Treated as a deductible fee in this study since it reduces what the fleet/driver ultimately retains from the fare. Negative values are clipped to 0 during ABT construction.	0 or 0.50 for almost all trips (36 trips up to 22.14) (USD)
tip_amount	Float	Gratuuity paid by the passenger. Recorded for card and app payments (credit card and flex-fare trips); cash tips are not captured by the system and will appear as 0 even if a cash tip was given. Not included in the total_fees deduction, since a tip is earned income rather than a cost.	0.0 to 550.00 (USD)
tolls_amount	Float	Total toll charges incurred during the trip (e.g., bridges, tunnels). Not included in total_fees for this study, since tolls are typically passed through/reimbursed rather than a true revenue reduction for the fleet.	0.0 to 115.87 (USD)
improvement_surcharge	Float	A fixed regulatory surcharge assessed on hailed trips at flag drop. Included as one of the components summed into total_fees.	0, 0.3, or 1.0 (USD)
total_amount	Float	The full amount charged to the passenger for the trip, combining fare_amount, extra, mta_tax, tip_amount, tolls_amount, improvement_surcharge, Airport_fee, and the congestion-related fees. Does not include cash tips. This is the gross revenue figure before total_fees is deducted to arrive at net_revenue.	1.01 to 1,345.71 (USD)

Column / Variable Name	Data Type	Description / Meaning	Possible Values / Categories
congestion_surcharge	Float	A New York State-mandated surcharge applied to trips that begin, end, or pass through designated congestion zones in Manhattan. One of the five fee components summed into total_fees, and central to Analytical Question 2 regarding congestion-fee impact. Flex-fare trips record it as 0 in the source data; during ABT construction it was restored to 2.50 on the 591,029 trips whose total_amount was exactly 2.50 higher than the sum of their recorded components.	0.0 or 2.50 for almost all trips (3 trips at 0.75 or 1.00) (USD)
Airport_fee	Float	A flat fee applied specifically to pickups at LaGuardia (LGA) and John F. Kennedy (JFK) airports. Included in total_fees since it is a mandatory deduction tied to specific pickup zones. Blank for the 757,387 flex-fare trips in the source data and filled with 0 during ABT construction.	0.0 or 1.75 (USD)
cbd_congestion_fee	Float	The MTA's Congestion Relief Zone per-trip charge for operating within Manhattan's Central Business District, effective January 5, 2025. This is the specific 2025 policy change that motivates the entire study's business question, and is included in total_fees.	0.0 or 0.75 for almost all trips (13 trips between 0.25 and 1.50) (USD)
duration_min	Float	Total trip duration in minutes, calculated as the difference between tpep_dropoff_datetime and tpep_pickup_datetime. Used to compute speed_mph and net_revenue_per_hour. Not used as a predictor in any of the three models, since it would not be known before a trip occurs (data leakage risk); it enters the models only through the zone-hour target and the lagged, zone-level average speed features (avg_speed_mph_lag1h and avg_speed_mph_lag24h).	1.0 to 179.97 minutes
speed_mph	Float	Average speed for the trip, calculated as trip_distance divided by duration_min (converted to hours). Serves as the numeric basis for the congestion_level category and, at the aggregated zone-hour level, for the lagged congestion features (avg_speed_mph_lag1h and avg_speed_mph_lag24h) used in modeling.	1.0 to 80.0 mph
pickup_hour	Integer	The hour of day (24-hour format) extracted from tpep_pickup_datetime. One of the core time-based predictors, capturing peak vs. off-peak demand patterns.	0 to 23
day_of_week	String	The name of the day of the week on which the trip started, derived from tpep_pickup_datetime. Captures weekly demand and revenue patterns (e.g., weekday commute vs. weekend leisure travel). It is one-hot encoded for SVR and the Neural Network, and used as a native categorical feature by XGBoost.	Monday, Tuesday, Wednesday, Thursday, Friday, Saturday, Sunday
is_weekend	Boolean	A simplified flag derived from day_of_week, set to true for Saturday and Sunday. Provides the model a more direct weekday/weekend signal than day_of_week alone.	True, False
pickup_year	Integer	The calendar year extracted from the pickup date. Used only to confirm that the ABT correctly represents the intended 2025 study period (post-congestion-fee implementation); not used as a model predictor since only one year is present in the data.	2025
total_fees	Float	Engineered column: the sum of cbd_congestion_fee, congestion_surcharge, Airport_fee, mta_tax, and improvement_surcharge. Any individual negative fee value is clipped to 0 before summing, so this column itself cannot be negative. Represents the total mandatory/regulatory deduction applied to a trip's gross revenue.	0.0 to 25.39 (USD)

Column / Variable Name	Data Type	Description / Meaning	Possible Values / Categories
net_revenue	Float	Engineered column, calculated as total_amount minus total_fees. Represents what is actually retained from the fare after regulatory and congestion-related deductions, before being converted to an hourly rate. One refund trip with a negative extra has a negative value; the modeling notebook keeps only trips with positive net revenue.	-6.89 to 1,340.96 (USD)
net_revenue_per_hour	Float	Trip-level net revenue per occupied driving hour, calculated as net_revenue / (duration_min / 60) and capped at the 99th percentile (220.83) during ABT construction to limit unusually short, high-fare trips. The target variable for all three models (XGBoost, SVR, Neural Network) is the zone-hour version of this measure, which the modeling notebook rebuilds from the uncapped totals: total net_revenue / total duration_min x 60 for each Pickup Zone x Date x Hour cell.	-38.22 to 220.83 (USD per occupied driving hour)
fee_impact_pct	Float	Engineered column expressing total_fees as a percentage of total_amount ((total_fees / total_amount) x 100). Directly supports Analytical Question 2, which examines how much of a trip's revenue is absorbed by congestion-related charges. Only one refund trip exceeds 100%.	0.0% to 137.24%
fare_tier	String	Categorical bucket grouping trips by total_amount into Low/Medium/High bands of equal size (quantile-based binning into tertiles during ABT construction). Used for describing trip-mix characteristics and for threshold-style findings under Analytical Question 2.	Low Fare (1.01 to 18.18), Medium Fare (18.19 to 26.94), High Fare (26.95 and above) (USD)
trip_length_tier	String	Categorical bucket grouping trips by trip_distance into Short/Medium/Long bands of equal size (tertiles). Used alongside fare_tier to characterize which kinds of trips remain profitable after congestion fees are applied.	Short Trip (0.11 to 1.33), Medium Trip (1.34 to 2.81), Long Trip (2.82 and above) (miles)
congestion_level	String	Categorical bucket derived from speed_mph, grouping trips into Light/Moderate/Heavy traffic bands (0 to 8 mph Heavy, above 8 to 15 mph Moderate, above 15 mph Light). Directly supports Analytical Question 3, which examines congestion tipping points where hourly revenue efficiency declines.	Light Traffic, Moderate Traffic, Heavy Traffic

Note on data leakage

Several fields above (duration_min, speed_mph, fee_impact_pct, and same-trip fare/distance figures) describe conditions that are only known after a trip is completed. For modeling, these are not used at the individual-trip level as predictors; instead, the trip-level data is aggregated into Pickup Zone x Date x Hour cells, and lagged (1-hour and 24-hour prior) zone-level trip counts, average speeds, and average fee impact are used as predictors, consistent with the no-leakage requirement in the FA2 methodology. Same-trip fare and distance figures are not used as predictors.